

Elton Chang

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EDUCATION

Carnegie Mellon University	May 2026
<i>M.S. in Data Analytics for Science, GPA:3.97/4.0</i>	Pittsburgh, PA
University of California, Santa Cruz	June 2025
<i>B.S. in Computer Science, GPA:3.7/4.0</i>	Santa Cruz, CA

EXPERIENCE

Founder/Technical Lead	August 2025-Present
<i>Ashe System</i>	Santa Cruz, CA
- Modeled \$15K/month in avoidable downtime savings per operator with more than 1.2K record risk engine that flags grounding risk, diagnoses root causes, and recommends preventive actions. - Defined and build more than 50 MVP requirements from validations with multiple cargo and small-aircraft owner-operators across maintenance, inventory, suppliers, certification, and scheduling.	

Undergraduate Research Assistant	September 2024-June 2025
<i>University of California, Santa Cruz</i>	Santa Cruz, CA
- Cut research data processing time from days to under an hour by building automated pipelines to clean, standardize, and structure 600 research files totaling 24GB. - Improved CNN-based image classification accuracy from 63% to 82% by optimizing model architecture, preprocessing workflow, and training configuration.	

Infrastructure Engineer Intern	July 2023-Sep. 2023
<i>Wintec Industries Inc.</i>	Newark, CA
- Reduced configuration drift across 15+ non-production hosts by implementing Infrastructure-as-Code provisioning for standardized testing environments. - Cut anomaly detection time by 40% by engineering telemetry, alerting, and reliability dashboards across enterprise backup and replication pipelines.	

Software Engineer Intern	July 2022-Sep. 2022
<i>Wintec Industries Inc.</i>	Newark, CA
- Built a cloud-hosted internal file management system with automated ingestion, metadata indexing, and lifecycle tracking for company data. - Improved onboarding and platform adoption by 35% by creating developer and user-facing documentation for architecture, workflows, and system usage.	

PROJECTS

Next Best Action (Collaboration with PNC)
- Ranked high-impact advisor opportunities with a PNC Next Best Action pipeline using HistGB, LightGBM, and LightGBM-XGBoost ensemble to predict next-month product adoption. - Result achieving 0.811 ROC AUC and 95% top-3 hit rate on supported recommendations.
Torflix (Open source at github)
A BitTorrent-backed movie library with a FastAPI service that owns swarm I/O, storage, and progressive streaming. With web clients available for catalog, magnet enqueue, and playback while transfers complete.
Quantifying Meritocracy (Research)
Extended the University of Catania's Talent vs. Luck model by introducing a solidarity parameter and applying statistical modeling with agent-based simulations to quantify systemic drivers of inequality.

SKILLS

- Python | TypeScript | Java | C | Go | SQL | R | React | Next.js | Vue | LangChain | LangGraph | LlamaIndex